OLYMPIAD · EVERYDAY MATHEMATICS · OLYMPIAD PREPARATION · SHORT NOTES

Chapter 15: Speed, Distance and Time

Topic-wise short notes with solved examples

The speed formula, changing units, average speed, relative speed, trains and boats in streams.

1 Speed, distance and time

- $\text{speed} = \dfrac{\text{distance}}{\text{time}}$, so $\text{distance} = \text{speed} \times \text{time}$ and $\text{time} = \dfrac{\text{distance}}{\text{speed}}$.
- Keep the units matching: km with hours gives km/h, and metres with seconds gives m/s.
- To change km/h to m/s, multiply by $\dfrac{5}{18}$. To change m/s to km/h, multiply by $\dfrac{18}{5}$.

Example 1.1 Change 72 km/h to m/s.

SOLUTION

$72 \times \dfrac{5}{18} = 20.$

20 m/s

Example 1.2 How long does a car take to cover 150 km at 50 km/h?

SOLUTION

$\text{time} = \dfrac{150}{50}.$

3 hours

2 Average speed

- **Average speed** $= \dfrac{\text{total distance}}{\text{total time}}$. It is **not** the average of the two speeds unless the times are equal.
- If the **same distance** is covered at speeds x and y, the average speed is $\dfrac{2xy}{x+y}$.

Example 2.1 A car travels at 60 km/h for 2 hours and at 40 km/h for 3 hours. Find the average speed.

SOLUTION

Distance $= 120 + 120 = 240$ km. Time $= 5$ hours. Speed $= \dfrac{240}{5}$.

48 km/h

Example 2.2 A person goes to a place at 40 km/h and returns by the same road at 60 km/h. Find the average speed for the whole journey.

SOLUTION

The distances are equal, so the average speed is $\dfrac{2 \times 40 \times 60}{40 + 60} = \dfrac{4800}{100}$.

48 km/h

3 Relative speed

- Two objects moving **towards each other** (or in opposite directions) have relative speed = the **sum** of the speeds.
- Two objects moving in the **same direction** have relative speed = the **difference** of the speeds.
- The time to meet or to catch up is $\dfrac{\text{gap}}{\text{relative speed}}$.

Example 3.1 Two cars start 300 km apart and move towards each other at 50 km/h and 70 km/h. When do they meet?

SOLUTION

Relative speed $= 50 + 70 = 120$ km/h. Time $= \dfrac{300}{120}$.

2.5 hours

4 Trains and boats

- A train of length L crossing a **pole** or a person must cover its own length L. Crossing a **platform** or bridge of length P it must cover $L + P$.
- A boat with speed u in still water in a stream of speed v goes **downstream** at $u + v$ and **upstream** at $u - v$.

Example 4.1 A train 150 m long runs at 54 km/h. How long does it take to cross a pole, and to cross a platform 350 m long?

SOLUTION

Speed $= 54 \times \dfrac{5}{18} = 15$ m/s.

Pole: $\dfrac{150}{15} = 10$ s. Platform: $\dfrac{150 + 350}{15} = 33\dfrac{1}{3}$ s.

10 s and $33\dfrac{1}{3}$ s

Example 4.2 A boat has a speed of 10 km/h in still water and the stream flows at 2 km/h. How long does it take to go 24 km downstream?

SOLUTION

Downstream speed $= 10 + 2 = 12$ km/h. Time $= \dfrac{24}{12}$.

2 hours

REMEMBER

Change every speed to one unit first, either km/h or m/s, before you divide.

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